

SUMMARY — SOCIAL MEDIA ENGAGEMENT ANALYTICS

Project Overview

This project analyzes 5,000 social media posts using Python, Pandas, NumPy, Matplotlib, Seaborn, and Plotly. The analysis focuses on understanding engagement patterns, audience behavior, content performance, sentiment, device usage, and posting time.

Key Findings

1. Best Performing Post Type:

Video posts achieved the highest average engagement score.

2. Best Performing Category:

Music was the highest-performing post category based on average engagement score.

3. Best Performing Country:

France recorded the highest average engagement score among the countries analyzed.

4. Sentiment Analysis:

Negative sentiment posts recorded the highest average engagement score and engagement rate in this dataset.

5. Age Group Analysis:

Users below 18 showed the highest average engagement score.

6. Device Usage:

Mobile users recorded the highest average watch time.

7. Best Posting Time:

Posts published around 12:00 AM (midnight) recorded the highest average impressions.

8. Engagement Analysis:

Likes, comments, and shares were combined to create an engagement score for comparing the performance of different posts.

Conclusion

The analysis shows that video and music content performed strongly in terms of engagement. Audience characteristics also influenced engagement, with users below 18 showing higher interaction levels. Mobile devices were associated with the highest average watch time, highlighting the importance of mobile-friendly content. The dataset also showed higher engagement for negative sen